Yonghui Liu

Ph.D. Candidate, KAIST
liuil@kaist.ac.kr | lyh360121.github.io

Education

KAIST Aug 2023 - Present
Ph.D. candidate, Cho Chun Shik Graduate School of Mobility; advisor: Prof. Inhi Kim

Doctoral thesis: *Compression-Reconstruction Co-Design for Urban Mobility Trajectories* (in progress).

Zhejiang University (ZJU) Sep 2018 - Mar 2021
M.S., Ocean Engineering

Nanchang University (NCU) Sep 2014 - May 2018
B.S., Physics

Research Experience

Doctoral Research, KAIST Aug 2023 - Present
Research on trajectory observation-reconstruction co-design, with published studies in trajectory modeling and road-network representation learning.

Efficient trajectory modeling

- **ProChunkFormer.** Developed a progressive chunked Transformer for sparse GPS reconstruction. On Porto taxi data at 240-second sampling intervals, reduced runtime by 20.7% and road-network MAE by 25.1% versus an autoregressive Transformer. [2]
- **SDA.** Designed a controlled-gating protocol to quantify history, state, and goal-prior contributions in LSTM/Transformer forecasting on Porto taxi and ETH pedestrian data. [1]

Representation learning with language and graph models

- **Road-network representations.** Combined LLM-derived semantics with heterogeneous graph learning. Achieved 0.6766 F1 in Porto-to-Lisbon transfer without retraining, versus 0.6968 F1 for a Lisbon-trained model. [6]
- **Cross-taxonomy alignment.** Combined LLM pseudo-labels with structural embeddings. With 1% human-labeled data, achieved 51.3% accuracy versus 40.0% for direct supervised training. [5]

Current Research Projects

Route-Conditioned Traffic Simulation *Ongoing research*
- Developing route-conditioned traffic generation on SMART/CAT-K by jointly fine-tuning a route encoder, conditioning branch, and motion decoder.
- Implemented recorded-scene initialization and offline route allocation under feasibility and quota constraints; route-control evaluation and integration are ongoing.

Autonomous Traffic Sensor Web (ATSW) *Working research prototype*
- Implemented a source-receiver replay system with adaptive GPS reporting and Kalman estimation, supporting policy comparison using position error and report counts.
- Built a FastAPI backend and React/TypeScript interface for inspecting receiver trajectories, reporting decisions, and historical reconstruction.

Publications

Journal Articles

1. **Yonghui Liu** and Inhi Kim. Sequential decomposition and attribution for trajectory forecasting. *Knowledge-Based Systems*, 335:115204, 2026.

2. **Yonghui Liu**, Qian Li and Inhi Kim. Enhanced trajectory reconstruction from sparse and noisy GPS data: A progressive chunked transformer approach. *Communications in Transportation Research*, 5:100200, 2025.

3. Hulong Zeng, Pengcheng Jiao, **Yonghui Liu**, Ujjwal Karki and Fang Z. Peng. Analytical modeling of resonant converters in frequency domain for wireless power transfer: Continuous current mode (CCM) operation. *Energy*, 246:123371, 2022.

4. Thomas Pähtz, **Yonghui Liu**, Yuezhang Xia, Peng Hu, Zhiguo He and Katharina Tholen. Unified model of sediment transport threshold and rate across weak and intense subaqueous bedload, windblown sand, and windblown snow. *Journal of Geophysical Research: Earth Surface*, 126(4):e2020JF005859, 2021.

Conference Papers

5. **Yonghui Liu** and Inhi Kim. Cross-Taxonomy Label Alignment in Transportation Networks With Prompt-Driven Large Language Models. *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, pp. 1724-1729, 2025.

6. **Yonghui Liu** and Inhi Kim. A Generalizable Semantic Graph Learning Framework for Road Network Representation. *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, pp. 720-725, 2025.

Professional Experience

Software Engineer May 2021 - Aug 2023
Meituan

AI Research Intern Apr 2020 - Oct 2020
Advanced Institute of Information Technology, Peking University

Technical Skills

Methods: Transformers, GNNs, LLM-based semantic supervision, conditional generation, Kalman filtering.

Tools: Python, PyTorch, FastAPI, React, TypeScript, SQLite.

Languages

Chinese: Native; **English**: Fluent; **Korean**: Basic